

Optical Character Recognition in Android Using Tesseract



This article, which is aimed at Android developers and image processing enthusiasts, explains how to extract text out of a captured image, using the Tesseract library.

Optical character recognition (OCR) is a technology that enables one to extract text out of printed documents, captured images, etc. Android currently doesn't come prebundled with libraries for OCR, unlike for voice-to-text conversion, which can be done using *android.speech.RecognizerIntent*. To build an Android app that can perform OCR or leverage these capabilities, one might have to opt for external libraries.

About Tesseract

Tesseract is a well-known open source OCR library that can be integrated with Android apps. It was originally developed by Hewlett Packard Labs and was then released as free software under the Apache licence 2.0 in 2005. The development has been sponsored by Google since 2006.

One could leverage OCR for building many Android applications like, for instance, scanning business cards and importing the text to a contact list, scanning an image and playing aloud the extracted text for the visually impaired, and so on. In this article, we will use the tess-two library, which is Tesseract with a Java native interface layer over it, to compile on Android platforms.

Integration with Android

The prerequisite is that the device should be running Android 2.3 (Honeycomb) or a higher version. In Android Studio, which is the official IDE for Android app development, tess-

two could be integrated in the following way:

1. Start with downloading tess-two from <https://github.com/rmtheis/tess-two>.
2. Once complete, create a folder named 'libraries' in your application directory. Place the downloaded tess-two folder under *Libraries*. We may have to do this as tess-two needs to be configured using Gradle. The folder names here could be anything, but the same name needs to be given in *settings.gradle* and the app's *build.gradle* while building the app.
3. Create *build.gradle* under the tess-two directory to configure the script that will build tess-two. You could use this simple *build.gradle* code given below:

```
apply plugin: 'com.android.library'
```

```
buildscript {
    repositories {
        mavenCentral()
    }
}
```

```
android {
    compileSdkVersion 23
    buildToolsVersion "23.0.3"

    defaultConfig {
```